

Training Configuration and Summary:

Sequencing:

- timesteps: 30
- batch_size: 128

Model Hyperparameters:

- min_units: 50
- max_units: 150
- units_step: 50
- dropout_min: 0.2
- dropout_max: 0.3
- dropout_step: 0.1
- activation_functions: ['tanh']
- optimizers: ['adam']

Tuning:

- max_epochs: 50
- hyperband_iterations: 3

BiLSTM Multi-Step Training Report

Report Generated: 2025-05-02 20:47:38

Training Start: 2025-04-27_09-38-50

Training End: 2025-04-27_10-08-17

Training Duration: 29.44 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: adam

Dropout Rate: 0.20

MAE: 1.1405

MSE: 2.1211

RMSE: 1.4564

R2 Score: 0.6494

Multi-Step Forecast:

Forecast Steps: 10

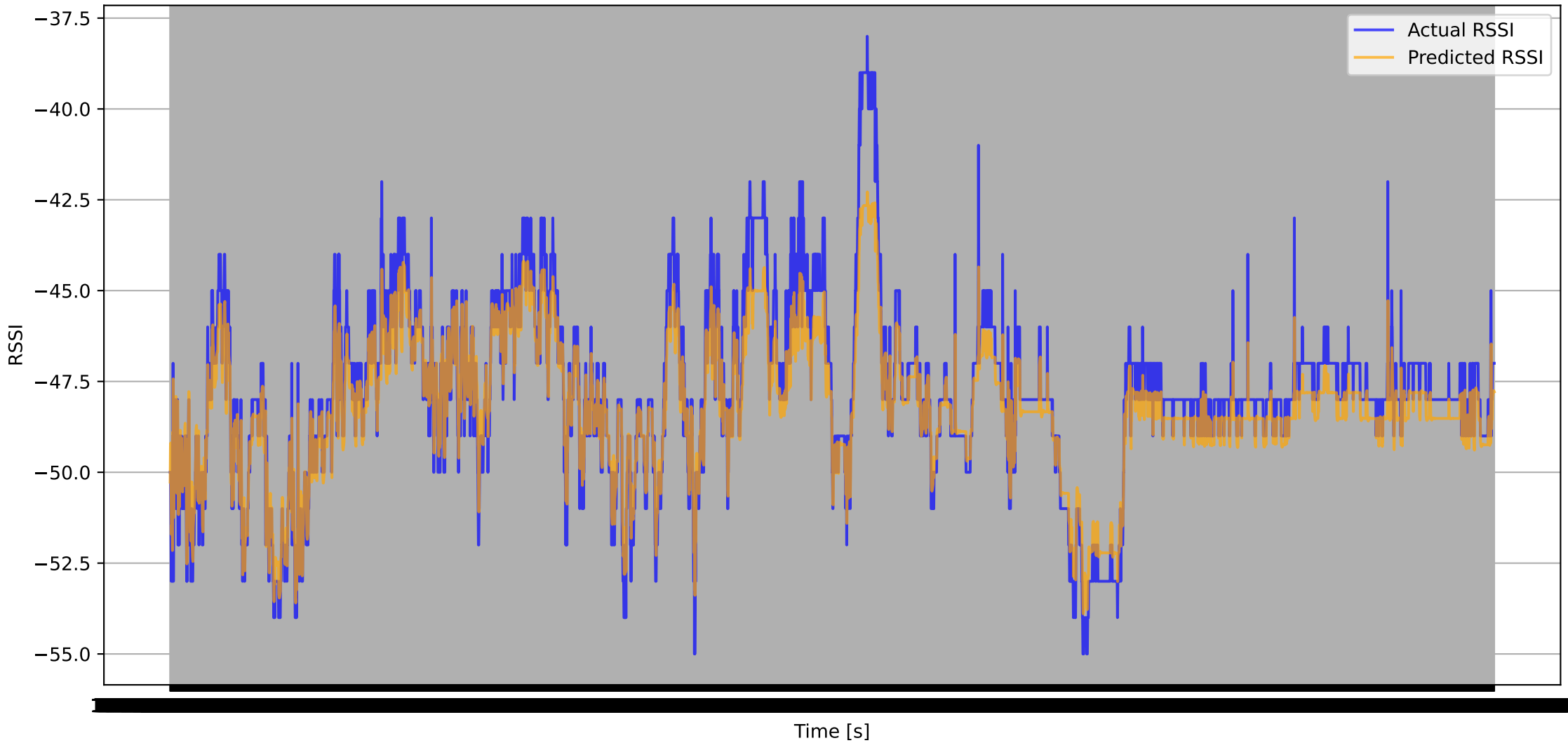
MAE: 0.0246

MSE: 0.0011

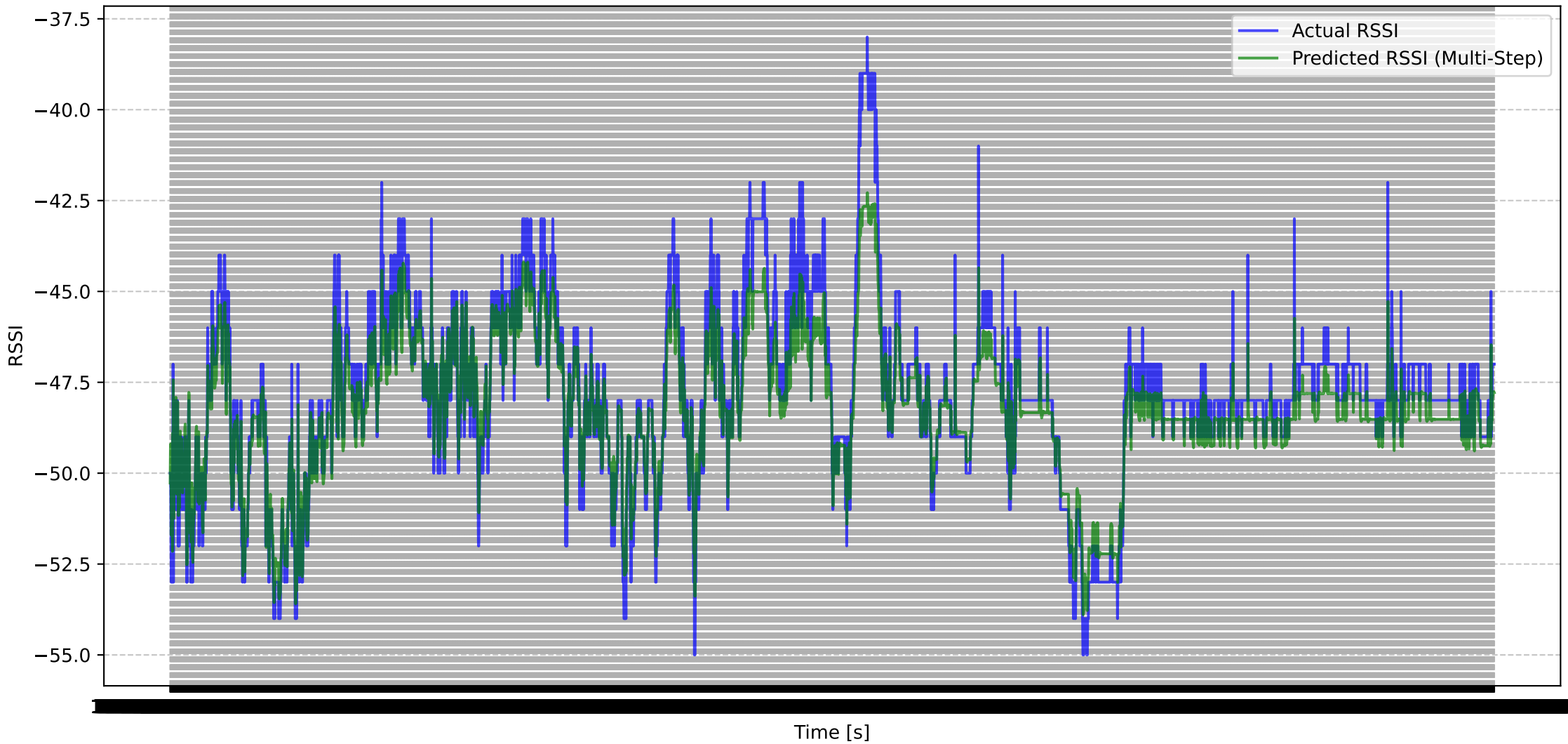
RMSE: 0.0330

R2 Score: 0.8489

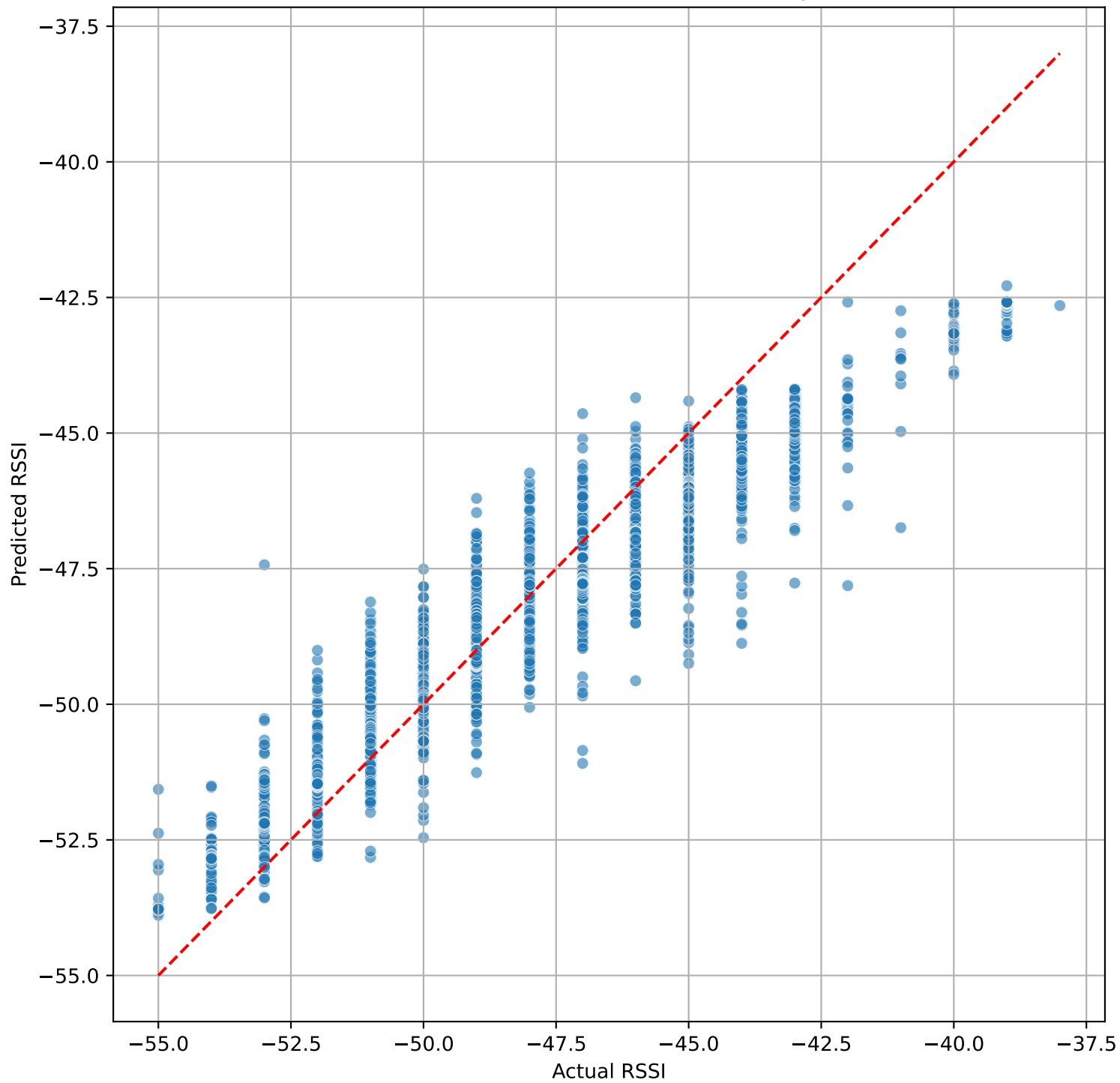
Actual vs Predicted RSSI (First Step)



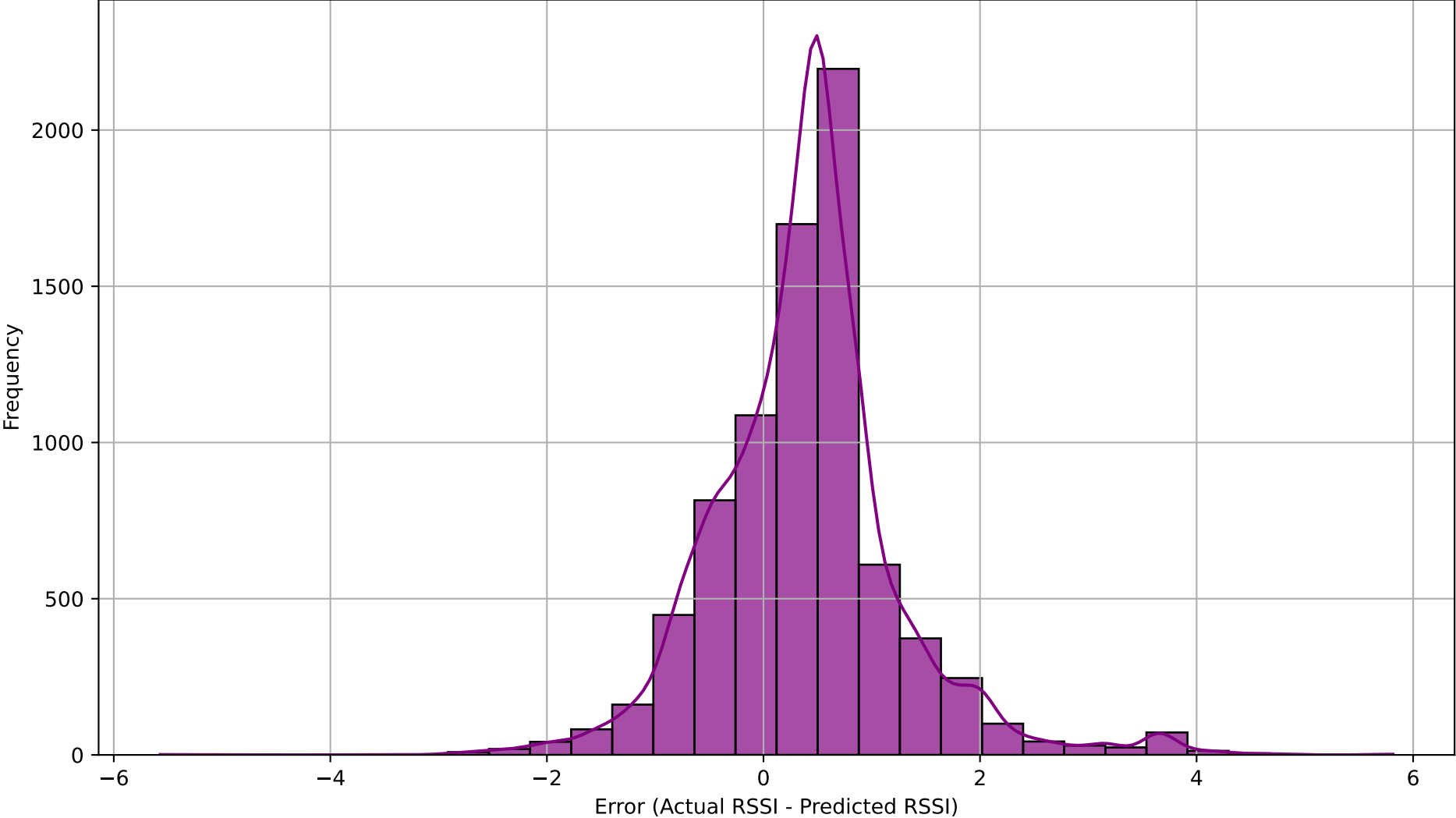
Actual vs Predicted RSSI (First Step vs Multi-Step)



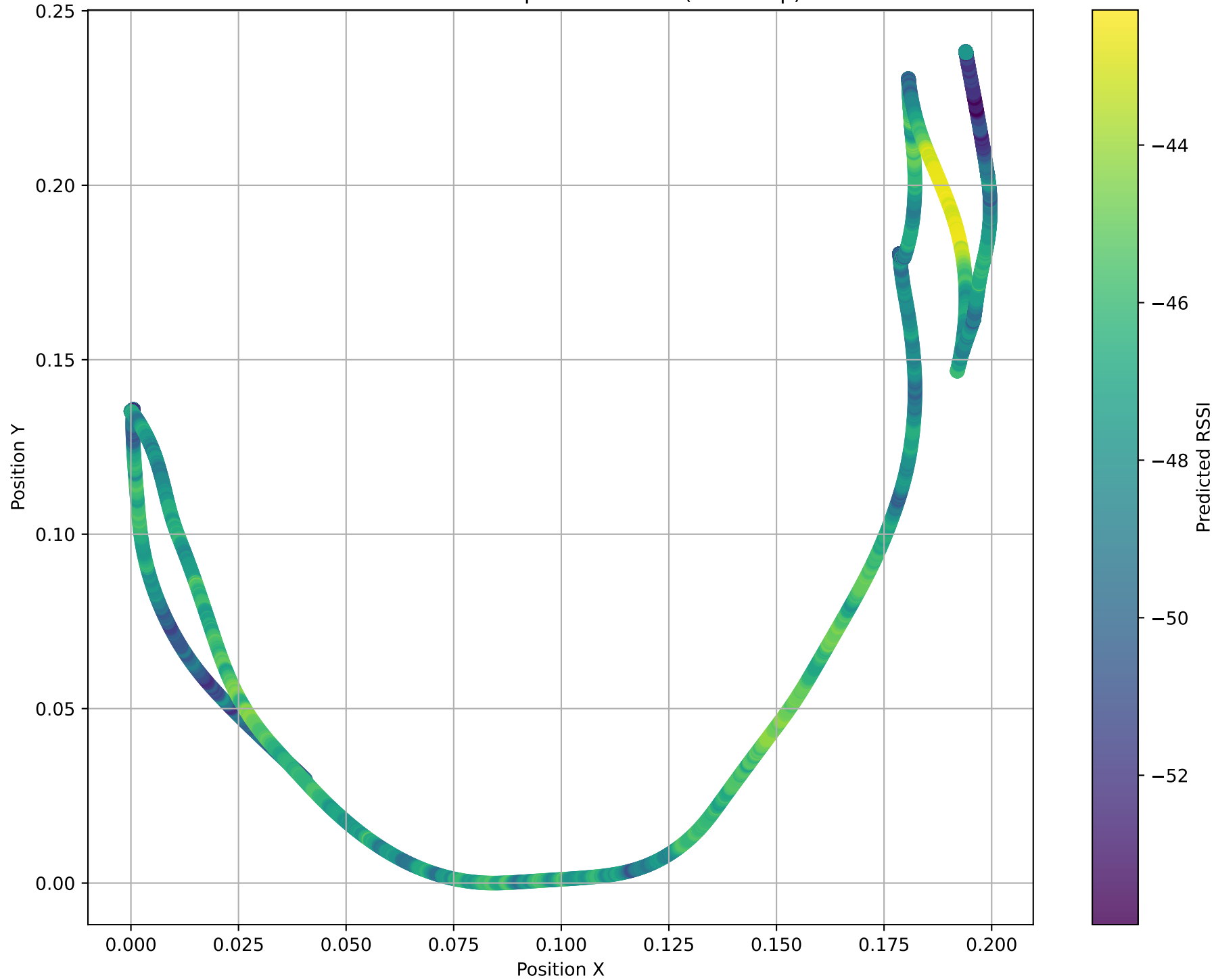
Actual vs Predicted RSSI (First Step)



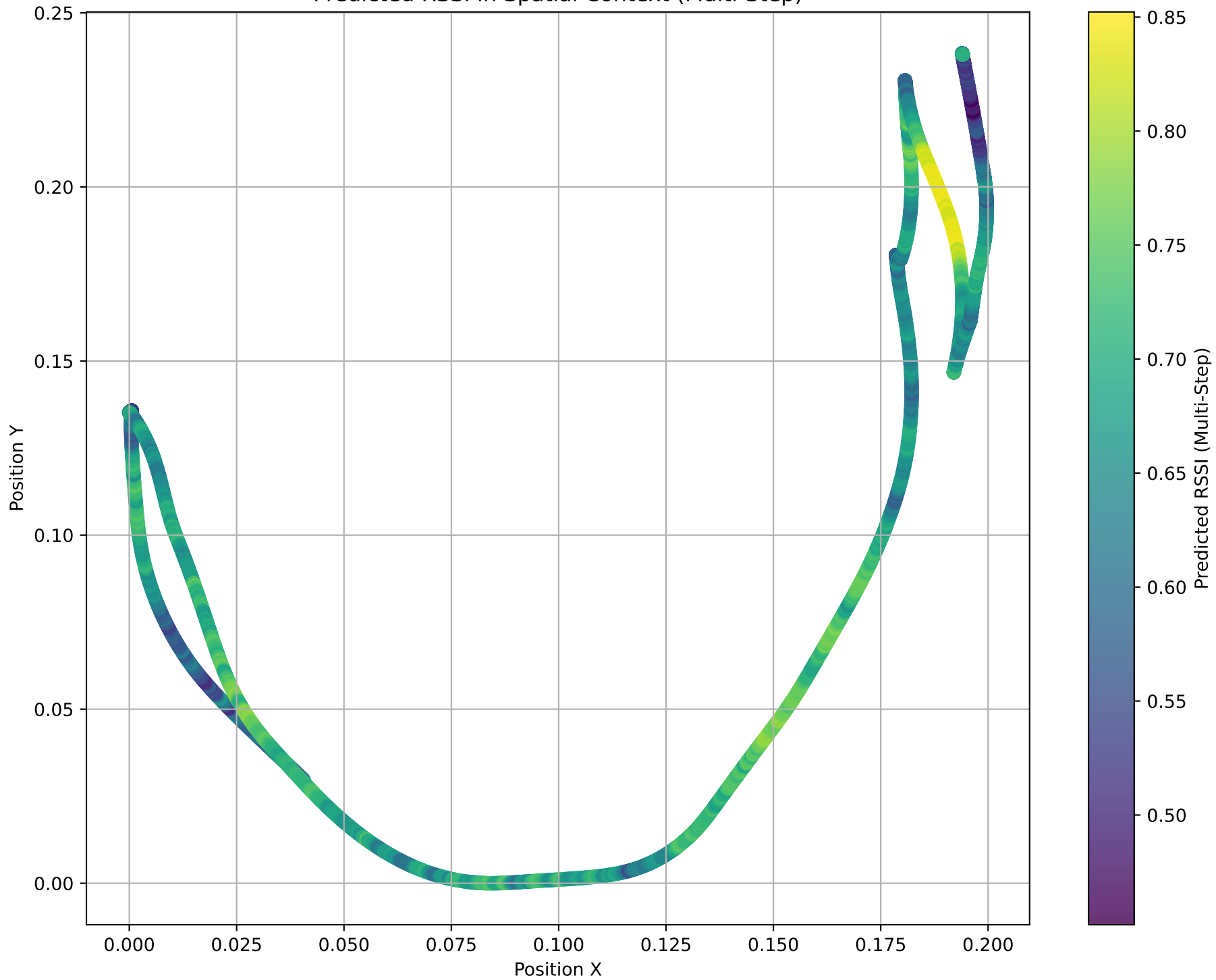
Distribution of Prediction Errors (First Step)



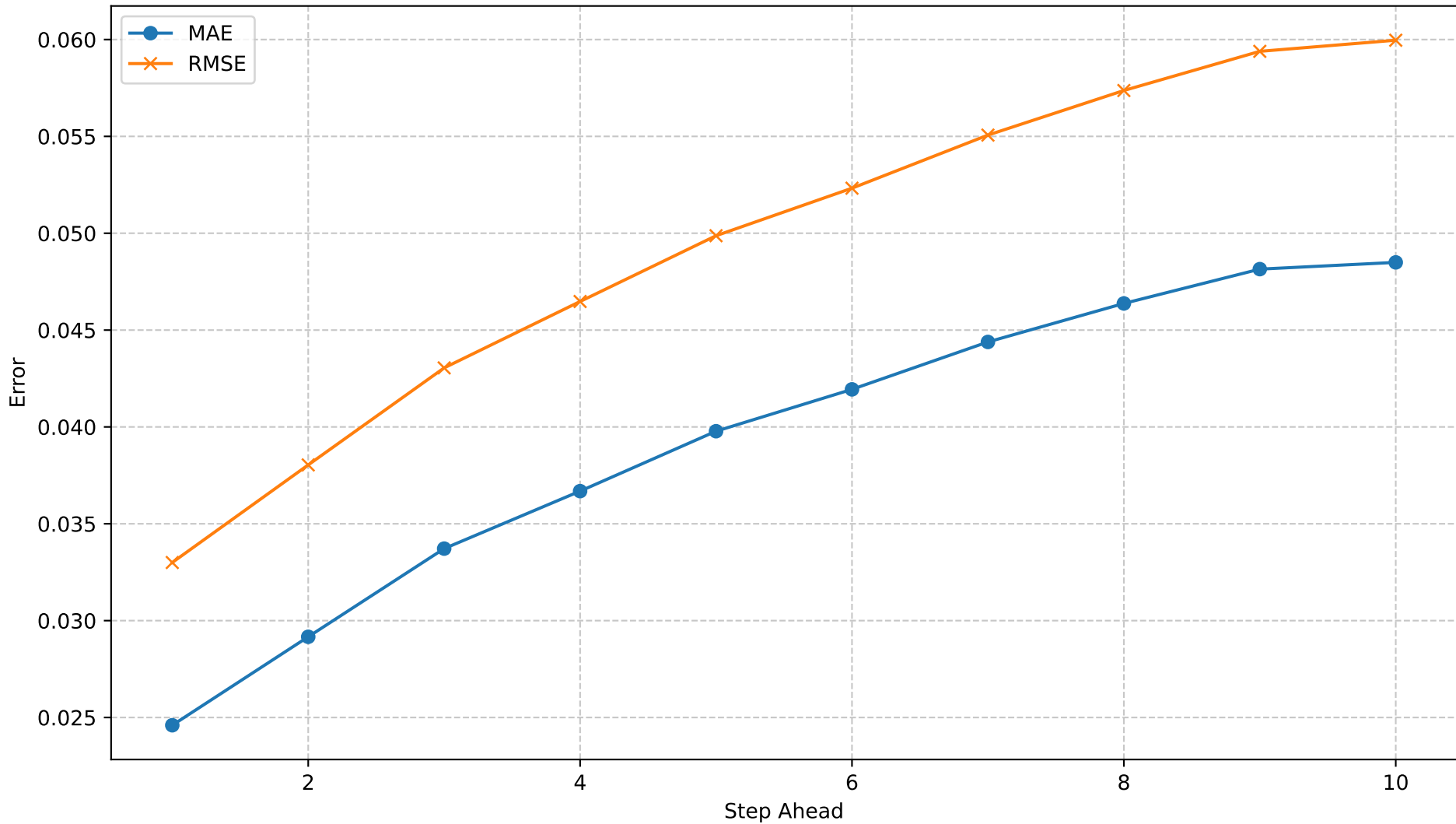
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step)



Multi-Step Forecast Error by Step



Resumo Automático:

- Forecast Steps configurados: 10
- Passos com MAE $\leq$ 2.0: 10 passo(s)
- MAE no último passo: 0.0485

[V] O modelo mantém boa acurácia durante quase toda a sequência de previsões.